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In [13]:
```
from sklearn import preprocessing
import numpy as np
```

Carga de datos.

In [14]:
```
X_train = np.array([[ 1., -1., 2.],
                    [ 2., 0., 0.],
                    [ 0., 1., -1.]])
```

Limpieza de datos: estandarización.

In [15]:
```
standardizer = preprocessing.StandardScaler()
```

Estandarizamos los datos de entrenamiento

In [16]:
```
X_train_std=standardizer.fit_transform(X_train)
print(X_train_std)
```

```
[[ 0.         -1.22474487  1.33630621]
 [ 1.22474487  0.         -0.26726124]
 [-1.22474487  1.22474487 -1.06904497]]
```

In [17]:
```
X_train_2 = np.array([[ 1., -1., 2.],
[ 2., 0., 0.],
[ 0., 1., -1.]])
```

Aplicamos los métodos "fit" y "transform" por separado

In [18]:
```
f=preprocessing.StandardScaler()
f=f.fit(X_train)
X_train_2_std = f.transform(X_train_2)
print(X_train_2_std)
# ???
```

```
[[ 0.         -1.22474487  1.33630621]
 [ 1.22474487  0.         -0.26726124]
 [-1.22474487  1.22474487 -1.06904497]]
```

Comprobar si las matrices estandarizadas son iguales

In [19]:
```
iguales = np.allclose(X_train_std, X_train_2_std)
print('¿Son las matrices iguales?:', iguales)
```

```
¿Son las matrices iguales?: True
```

GITHUB

- https://github.com/Failedbanjo/machine_learning_2

In [ ]:
```
```